

Regularized Exploratory Factor Analysis as an Alternative to Factor Rotation

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Abstract: Exploratory factor analysis (EFA) is widely used in psychological (assessment) research. Due to its exploratory nature, several researcher degrees of freedom exist on how to conduct the analysis. While simulation studies can provide meaningful insights into which factor retention methods to use to determine the number of latent factors, or which estimation methods recover parameter values most precisely given certain data characteristics, the issue of rotational indeterminacy makes it very difficult to decide which rotation method to apply. An alternative to the two-stage approach of extracting factors and subsequently rotating them to foster interpretability is the so-called regularized EFA. In this paper, we contrast both approaches and demonstrate how regularized EFA can be applied. In doing so, we want to encourage researchers to try out the approach themselves and help them find a way of EFA that appears less arbitrary compared to classical factor rotation.

Keywords: exploratory factor analysis, regularization, factor rotation, rotational indeterminacy, psychological assessment



Developing measurement models for psychological concepts plays an important role in psychological assessment. Usually, exploratory factor analysis (EFA) is used as a first step to carve out latent structures underlying an item set. Its importance for theory formation and construct development can be demonstrated by the role it played in the research process that led to the big five trait taxonomy (John & Srivastava, 1999). Besides personality psychological research, EFA is very popular in different research areas, for example, organizational psychology (Smith et al., 1983), cognitive psychology (Ryan et al., 2003), or developmental psychology (Ronald et al., 2005) – just to name a few. Despite this popularity, there are still a lot of debates on how to properly apply EFA (e.g., Fabrigar et al., 1999; Goretzko et al., 2021). When conducting an EFA, a researcher has to make several decisions – how many factors to retain, which estimation method to use, and how to rotate the initial factor solution to increase interpretability.

To determine the number of factors, researchers can choose from a large variety of so-called factor retention criteria. While simple heuristics such as the Kaiser-Guttman

or eigenvalue-greater-one rule (Kaiser, 1960) are criticized due to their lack of accuracy (e.g., Fabrigar et al., 1999), simulation-based approaches such as parallel analysis (Horn, 1965), the comparison data approach (Ruscio & Roche, 2012), or machine-learning-based factor retention criteria (Goretzko & Bühner, 2020, 2022a), as well as revised versions of these simple heuristics (e.g., the empirical Kaiser criterion; Braeken & Van Assen, 2017), can be seen as state-of-the-art approaches. Several simulation studies (e.g., Auerswald & Moshagen, 2019) can provide guidance on which of these methods perform best under which data conditions.

When it comes to choosing an estimation method, overview articles (e.g., Fabrigar et al., 1999; Goretzko et al., 2021) and textbooks (e.g., Widaman, 2012) provide comprehensive comparisons of the different extraction methods – from historically popular approaches like principal axis factoring (PAF) and classical maximum likelihood-based estimation (MLE) to weighted-least-squares (WLS) approaches with mean and variance adjustments (WLSMV) that appear to be superior for discrete data (Barendse et al., 2015). While the choice of an appropriate estimation method is still debated (e.g., Fabrigar et al., 1999; Goretzko et al., 2021) and researchers might want to compare results among methods (Widaman, 2012), some guidelines exist to use which method (e.g., to rely on MLE for continuous data or categorical data with at least five categories, on

WLS/WLSMV for categorical data with fewer categories, and to use PAF only if MLE produces Heywood cases; Goretzko et al., 2021).

Choosing a method for factor rotation seems to be much more difficult and to some degree arbitrary and often leaves researchers feeling uneasy. In EFA, researchers are confronted with so-called rotational indeterminacy which means that factor solutions are only determined up to admissible, yet arbitrary rotations, and an infinite number of rotated factor structures will explain the data equally well (e.g., Fabrigar et al., 1999). In other words, EFA solutions are not unique – there are rather different factor solutions that can be obtained via factor rotation and which all fit the data equally well. Hence, it remains an ambiguous decision for a researcher on how to rotate the initial solution. This decision can only be supported by theoretical insights and clear ideas about how the factorial structure of a concept should look but there is no data-driven way to decide among rotation methods (e.g., Goretzko et al., 2021). Rohe and Zeng (2022) show that this rotational invariance only occurs for multivariate normal data and that factors that stem from non-normal distributions are partially identifiable, though. Their proposed combination of principal component analysis and Varimax rotation to identify the factors appears to be especially useful for sparse data, but does not seem to be the best way for typical psychological data (Pargent et al., 2023).

Regularized exploratory factor analysis (REFA) might be a potential alternative to factor rotation that offers certain advantages and probably comes with less ambiguity for applied researchers (e.g., Scharf & Nestler, 2019). Therefore, we want to introduce the concept of REFA and regularization to EFA users in this paper. In doing so, we also want to guide them through the process of conducting REFA presenting exemplary code, and comparing the results of a REFA analysis with a “traditional” two-stage EFA (factor extraction + subsequent factor rotation).

Factor Rotation

Before we present regularization and REFA as an alternative to common EFA with factor rotation, we want to briefly recap the idea of rotating factor solutions for interpretability. Initial factor solutions are usually not easy to interpret since factors are ranked in descending order regarding the variance they are able to explain. This yields patterns where the first factors are associated with (almost) all variables and substantive cross-loadings are present. Through admissible rotations the orientation of the latent factors is changed, so that the complexity of the loading matrix is reduced (for a nice overview of rotation methods and a gentle introduction to the related concepts of simple structure and factor loading complexity; see Browne, 2001).

In general, oblique and orthogonal rotation methods are differentiated. Orthogonal rotations yield uncorrelated latent factors, while oblique techniques can provide solutions with substantially correlated factors. Many psychologists argue that oblique rotation methods should be preferred as they do not constrain between-factor correlations to zero which would be an unrealistic restriction, while others are in favor of orthogonal rotation because correlated factors would hamper interpretability (see, e.g., Fabrigar et al., 1999). However, it has to be stated that the decision between oblique and orthogonal rotation methods is solely a theoretical decision and cannot be informed by the observed data.

In psychological research, oblique rotation has gained popularity and is often preferred over orthogonal rotation methods (cf., e.g., Fabrigar et al., 1999; Goretzko et al., 2021). Promax (Hendrickson & White, 1964) and Oblimin (Clarkson & Jennrich, 1988) are the most prominent oblique rotation methods, while Varimax (Kaiser, 1958) is the most popular representative of orthogonal rotation (Goretzko et al., 2021). Oblimin is actually a family of different rotation criteria that can be selected by setting a parameter that controls the “obliqueness”, that is, the size of the between-factor correlations. Usually, the Quartimin criterion is the default solution for Oblimin in software solutions such as R (e.g., the psych package, Revelle, 2018) or SPSS (IBM Corp., 2019). Many of these popular rotation criteria can be formulated as special cases of a general rotation approach as described by Crawford and Ferguson (1970). Browne (2001) shows that different rotation criteria focus either on row-wise (i.e., the number of non-zero loadings per manifest variable) or column-wise (i.e., the number of non-zero elements per latent factor) complexity to obtain an interpretable solution (i.e., a pattern close to simple structure). He uses the so-called Crawford-Ferguson family (Crawford & Ferguson, 1970) to illustrate how different objective functions (“complexity functions”) can be used to foster interpretability through factor rotation.

This general complexity function of the Crawford-Ferguson family that is minimized in the rotation process weighs row-wise against column-wise complexity:

$$f(\Lambda) = (1 - \kappa) \sum_{i=1}^p \sum_{j=1}^k \sum_{l \neq j}^k \lambda_{ij}^2 \lambda_{il}^2 + \kappa \sum_{j=1}^k \sum_{i=1}^p \sum_{m \neq i}^p \lambda_{ij}^2 \lambda_{mj}^2, \quad (1)$$

with λ_{ij} being the loading of the i th manifest variable on the j th factor, p being the number of manifest variables, k being the number of latent factors, and κ being a constant that determines whether to focus on row-wise complexity (first part of the objective function) or the column-wise complexity (second part of the function). κ can take values between zero and one, with $\kappa = 0$ corresponding

to the Quartimin rotation criterion (i.e., one version of the Oblimin criterion; Clarkson & Jennrich, 1988) that focuses on row-wise complexity and $\kappa = 1$ corresponding to CF-Facparsim (Crawford & Ferguson, 1970) which tries to achieve a simple structure by maximizing factor parsimony.

Some simulation studies (e.g., Sass & Schmitt, 2010; Schmitt & Sass, 2011) try to derive recommendations for which rotation criteria perform well for different theoretical assumptions. However, those results are only useful in situations where researchers are able to make educated decisions based on substantive theory. If a respective theory provides enough information, a semi-exploratory rotation-to-target approach (Myers et al., 2015) might be suitable as well. In most cases, where EFA is used to explore the latent factor structure, such information is missing. Accordingly, several authors (e.g., Browne, 2001; Fabrigar et al., 1999; Goretzko et al., 2021; Preacher et al., 2013) suggest comparing different rotation methods – ideally by splitting the sample and investigating the robustness of the factor rotation.

Regularized Exploratory Factor Analysis (REFA)

Instead of estimating the parameters of an EFA model in one step (factor extraction) and subsequently performing factor rotation in a second step to ensure interpretability, REFA has been proposed repeatedly as an alternative that might be less ambiguous (e.g., Goretzko & Bühner, 2022b; Goretzko et al., 2021; Scharf & Nestler, 2019; Trendafilov & Adachi, 2015; Yamamoto et al., 2017). There are different ways to conduct REFA. Usually, regularization is directly incorporated into the estimation process by adding a penalty term to the objective function in maximum likelihood estimation (e.g., Hirose & Yamamoto, 2014, 2015; Jacobucci et al., 2016; Scharf & Nestler, 2019). Accordingly, the objective function takes the form:

$$F_{\text{REFA}} = F_{\text{MLE}} + \gamma \cdot P(\Lambda, \Phi), \quad (2)$$

where F_{MLE} is the objective function for common maximum likelihood estimation, γ is a hyperparameter that determines how strongly the penalty affects the estimation (i.e., the penalty weight) and $P(\Lambda, \Phi)$, is a function that penalizes high parameter values (in this case, high loadings Λ or between factor correlations Φ). By adding the latter part to the objective function, fewer non-zero parameters are preferred (i.e., a sparser solution) and parameter values representing non-substantial relationships among variables get shrunk toward zero. That way, REFA promises to yield more interpretable solutions that are close to a simple structure.

There are a lot of different ways the penalty term $P()$ can be designed to achieve this goal. Hirose and Yamamoto (2014, 2015) propose REFA with a minimax concave penalty (MC+; Zhang, 2010) that is able to handle large and comparably sparse loading patterns very well (i.e., it introduces less bias than other penalties in high-dimensional settings with very few substantial correlations). The MC+ penalty can actually be seen as a family of penalties. Besides γ , the hyperparameter that determines the severity of the penalty, a second hyperparameter δ influences the exact form of the penalty. This second hyperparameter can be interpreted as a way to control the level of sparsity and the rigor with which model parameters θ are shrunk toward zero:

$$\gamma \cdot P_{\text{MC+}}(\theta) = \gamma \left(|\theta| - \frac{\theta^2}{2\gamma\delta} \right) I(|\theta| < \gamma\delta) + \frac{\gamma^2\delta}{2} I(|\theta| \geq \gamma\delta), \quad (3)$$

with $I()$ being the indicator function. For meaningful values of γ (i.e., $\gamma > 0$) the choice of δ determines how strict parameter values are set to zero. Values approaching one, yield a so-called hard thresholding, which means that parameter values below a threshold are automatically set to zero, while $\delta \rightarrow \infty$ leaves us with soft thresholding which means that some parameter values become zero, while others are shrunk towards zero (Hirose & Yamamoto, 2015).

This soft threshold operator is known as the least absolute shrinkage and selection operator (LASSO; Tibshirani, 1996) and is frequently used in regression models and machine learning applications (Tibshirani, 2011). It applies the L_1 -norm of the parameter vector θ as the penalty term:

$$\gamma \cdot P_{\text{LASSO}}(\theta) = \gamma \cdot \|\theta\|_1 = \gamma \cdot \sum_j |\theta_j|, \quad (4)$$

which is simply the sum of the absolute values of all parameters. Hence, it is easy to see that many non-zero elements and large parameter values are penalized and sparser solutions are preferred (i.e., loading patterns with fewer cross-loadings in the context of REFA).

An alternative to LASSO is the Ridge penalty (Hoerl & Kennard, 1970) which uses the L_2 -norm – which means the sum of the squared parameter values – as the penalty term:

$$\gamma \cdot P_{\text{Ridge}}(\theta) = \gamma \cdot \|\theta\|_2 = \gamma \cdot \sum_j \theta_j^2. \quad (5)$$

By using the L_2 -norm instead of the L_1 -norm, the Ridge penalty is able to shrink parameters toward zero but will not end up setting parameters to exactly zero. This variable selection property of LASSO (for a discussion, see also

Tibshirani, 2011) appears to be quite appealing to researchers, as psychologists usually favor models without any cross-loadings, that is, measurement models with many parameters constrained to be exactly zero (e.g., Hopwood & Donnellan, 2010).

In prediction settings, a combination of LASSO and Ridge tends to provide the best results. The more flexible ElasticNet penalty (Zou & Hastie, 2005) also has been proposed for REFA and showed quite promising results in the first simulation studies (Scharf & Nestler, 2019). Similar to the MC+ penalty, an additional hyperparameter α determines the mixture ratio of the L_1 and L_2 penalty:

$$\gamma \cdot P_{\text{Enet}}(\theta) = \gamma \cdot [(1 - \alpha)\|\theta\|_1 + \alpha\|\theta\|_2]. \quad (6)$$

For $\alpha = 0$, the ElasticNet penalty equals the LASSO approach and for $\alpha = 1$, it becomes a Ridge penalty. Hence, the smaller α , the more parameters will be shrunk to exactly zero. In general, the choice of the penalty will influence the sparsity of the factor solution, so choosing a penalty function also requires some theoretical assumptions which are similar to those a researcher has to make when applying factor rotation. The MC+ penalty, for example, is able to produce much sparser solutions than the LASSO approach, which seems to be fitting for contexts with very large item sets and few substantial correlations. Accordingly, REFA with MC+ might be the right choice when working with large-scale data (e.g., fMRI data, genome data, or large panel data). However, when it comes to questionnaire data and typical data sources in psychological assessment, the ElasticNet (and therefore LASSO and Ridge) penalty seems to be more appropriate, as the number of items subjected to an EFA is usually comparably small (e.g., $p = 16$ as in the data example) and loading matrices often deviate from a perfect simple structure (e.g., Scharf & Nestler, 2019). Selecting the penalty weight γ can still be seen as a researcher's degree of freedom, but in contrast to factor rotation where no data-driven way exists to decide between rotation methods, it can be determined using information criteria such as the BIC (e.g., Jacobucci et al., 2016).

REFA Demonstration on Simulated Data

To demonstrate the nature of REFA, we simulate data that are prototypical for psychological research and the application of EFA for psychological assessment. In this simplified exemplary scenario, we look at two latent factors that are represented by six items each. We assume the following loading matrix on the population level:

$$\Lambda = \begin{bmatrix} 0.6 & 0 \\ 0.6 & 0 \\ 0.6 & 0 \\ 0.6 & 0 \\ 0.6 & 0 \\ 0.6 & 0 \\ 0 & 0.6 \\ 0 & 0.6 \\ 0 & 0.6 \\ 0 & 0.6 \\ 0 & 0.6 \\ 0 & 0.6 \end{bmatrix}, \quad (7)$$

and a between-factor correlation of $\rho = 0.3$. Using the *myt-norm* package (Genz et al., 2018), we simulate a sample of 400 observations from this population and apply REFA with a LASSO penalty. It is able to estimate the true loading parameters (and between-factor correlations, $\hat{\rho}_{12} = 0.28$) well (see Table 1). Note that the ordering of factors is different from the data generating model. The parameter estimates, however, are considerably close to the truth.

In this example, you can see that regularization achieves the same goal as factor rotation in the common two-step approach to EFA (for a mathematical description of how regularization can be used to circumvent the issue of rotational indeterminacy; see Hirose & Yamamoto, 2014, 2015). The resulting loading matrix is quite sparse and close to a simple structure (i.e., close to the true pattern in the data-generating process). Hence, this approach provides similar results to an EFA with an oblique factor rotation that aims at minimizing row-wise complexity or item-wise complexity such as Oblimin (see Table 1).

Table 1. Estimated loading matrix for simulated example using REFA with LASSO penalty compared to EFA with Oblimin rotation

REFA		EFA + Oblimin	
Factor 1	Factor 2	Factor 1	Factor 2
−0.02	0.62	−0.04	0.63
0.01	0.65	0.00	0.66
0.05	0.54	0.04	0.54
0.06	0.57	0.05	0.58
0.02	0.60	0.01	0.61
0.00	0.58	−0.01	0.59
0.67	0.00	0.67	−0.02
0.64	0.01	0.65	0.00
0.63	0.06	0.63	0.05
0.62	−0.10	0.63	−0.12
0.54	0.06	0.55	0.05
0.60	0.10	0.60	0.09

Note. The factor correlations between both factors are estimated to be 0.28 (REFA) and 0.32 (EFA + Oblimin).

Data Example

For a step-by-step presentation of REFA, we analyze data from a German adaptation of the Prosociality Scale (Fassbender & Luhmann, 2021) that can be downloaded from the authors' open science framework repository (<https://osf.io/rp8au>). Their data are especially useful for our purpose as the number of variables is rather small ($p = 16$) which helps us to present factor loadings in a concise way. Besides, the factor structure appears to be deviating from a simple structure which may help demonstrate the differences among different penalties within REFA. The authors suggested a general factor and three residual factors in a confirmatory model which we do not want to pursue as we focus on EFA applications that rely on the common factor model (Fassbender & Luhmann, 2021). We use the full sample of $N = 446$ observations. Further information about data collection, exclusion criteria for the original study, and the analysis scripts of the original study can be found together with the data in the OSF repository of Fassbender and Luhmann (2021).

```
# the data can be downloaded from the OSF repository:
# https://osf.io/rp8au
# load the data and select all columns that should be
# subjected to REFA

data =
  read.csv2("Prosociality Data.csv")[, 4:19]
```

Regularized EFA in R

First, we load the *regsem* (Jacobucci et al., 2021), *penfa* (Geminiani et al., 2021b), and *IsIx* (Huang, 2020a) packages that can be used to perform regularized structural equation modeling. We will showcase how these three packages can be used for REFA as they provide different functionalities and penalties that are interesting for researchers in psychology. Other packages such as *fanc* (Hirose et al., 2022), for example, focus on specific penalties (MC+ in this case) and/or on specific models (e.g., EFA models). The discussed packages can be used more broadly (e.g., for structural equation modeling [SEM] in general or in the case of *IsIx* and *penfa* for specific multi-group scenarios). We will briefly discuss the difference and similarities between the three packages at the end of the paper.

Since LASSO, Ridge, and ElasticNet seem to be suitable choices for the penalty in psychological research settings, and due to their compatibility with the most popular package for latent variable modeling, the *lavaan* package (Rosseel, 2012), these packages seem to be a good choice

for REFA. Scharf and Nestler (2019) relied on *regsem* in their proof-of-concept study and contributed to the functionality of the package to make it more useful for exploratory approaches (i.e., for REFA), so we will first have a look at *regsem*. As the presented code briefly introduces different implementations and every time tries out many values for the penalty weight γ , we also provide exemplary code with a reduced grid in the Electronic Supplementary Materials, ESM 1, for readers who want to quickly check how the method works.

The *regsem* Package

```
library(regsem)
# package to perform regularized SEM and REFA

set.seed(31923)
# set seed for reproducibility of results
```

Before conducting an EFA, one has to determine the number of factors that should be retained. For this demonstration, we decided to use parallel analysis (Horn, 1965) as implemented in the *psych* package in R (Revelle, 2018). Of course, more modern approaches (Braeken & Van Assen, 2017; Goretzko & Bühner, 2020; Goretzko et al., 2021; Goretzko & Ruscio, 2023; Ruscio & Roche, 2012) might yield different results, but we do not aim at interpreting our results on a substantive level, so we just use one factor retention criterion.

```
library(psych)
# package for common EFA and to perform parallel analysis

fa.parallel(data, plot = FALSE, fa = "fa")$nfact
# 3 factors
```

As parallel analysis suggests three factors, we will conduct a REFA with three latent factors. A three-factor solution was also considered by Fassbender and Luhmann (2021) in their supplemental material. They also discuss other factor solutions and found a four-factor model or rather a model with a general factor and three fits best. Hence, readers interested in the measurement of prosociality and thorough discussion of different factor solutions are referred to the original article by Fassbender and Luhmann (2021).

In the first step of our analysis with the *regsem* package, we have to define the model in *lavaan* syntax. Depending on the number of manifest variables and latent factors, that can be quite tedious. Hence, we will use the *efaModel* function of the *regsem* package (based on the work of

Scharf & Nestler, 2019) that does this automatically. However, if researchers want to specify the model themselves because more specific models should be fitted, this can be done using the *lavaan* model syntax. Afterward, we specify some properties of the model (e.g., that all variables, manifest and latent, should be standardized to obtain standardized parameter estimates as usual in EFA).

```
# the function takes the number of latent factors and the
# names of all
# indicators as input

model = efaModel(nFactors = 3, variables =
  colnames(data))
regsem_mod = sem(model, data = data,
  int.ov.free = FALSE,
    int.lv.free = FALSE, std.lv = TRUE,
    std.ov = TRUE,
    auto.fix.single = FALSE, se = "none")
```

Within the *cv_regsem* function, it is possible to solely penalize the loadings λ_{ij} , all loadings, and between-factor correlations altogether, or only selected parameters. The latter can be done by directly providing indices that indicate which model parameters to penalize. We use the Bayesian information criterion (BIC) to find the optimal value for the penalty weight (note: within the function, the penalty weight is called *lambda* which should not be confused with the standardized loading parameters that we usually call *lambda*; the *lambda* in this function is what we have called γ in the description of the different penalties above). The *cv_regsem* may run quite a while depending on model complexity, sample size and the grid size for the penalty weight (i.e., how many γ values are tried out). Users might want to set the argument *verbose* to *TRUE* for a progress bar.

```
# Code from Scharf & Nestler (2019) to gather the indices of
# all loading parameters as well as all unique
# between-factor correlations to be able to penalize them

# Loadings = extractMatrices(regsem_mod)$A
# [extractMatrices(regsem_mod)$A>0]
Cors =
  unique(extractMatrices(regsem_mod)$S[
    grepl("f", colnames(extractMatrices(
      regsem_mod)$S)) &
    extractMatrices(regsem_mod)$S>0])

# Option 1: Lasso penalty for all loading parameters
# (factor correlations will not be included in the penalty term)
```

```
#
# results_loadings = cv_regsem(model = regsem_mod,
# pars_pen = "loadings",
#   mult.start = TRUE, multi.iter = 10, metric = "BIC",
#   n.lambda = 200, type = "lasso", jump = 10^-5,
#   lambda.start = 0.0001)

# Option 2: LASSO penalty for both loadings and factor
# correlations

# 200 penalty weights are tried, starting with 0.0001 and
# increasing this
# value in steps of 0.00001
# these values might have to be adjusted depending on the
# model size
# verbose = TRUE prints a progress bar

results_both = cv_regsem(model = regsem_mod,
  pars_pen = c(Loadings, Cors),
  mult.start = TRUE, multi.iter = 10,
  metric = "BIC",
  n.lambda = 200, type = "lasso", jump = 10^-5,
  lambda.start = 0.0001, verbose = FALSE)
```

We now can compare the results of the best solution according to the BIC ($\gamma = 0.00094$) with the solution produced by the largest penalty weight tested in our analysis ($\gamma = 0.00209$).

```
# loading parameters of best solution
loads_best = results_both$final_pars[Loadings]
# loading parameters of solution with largest penalty weight

loads_max = results_both$parameters[200,
  Loadings]
```

Table 2 shows the factor solution selected by BIC in comparison to a solution based on the largest penalty weight in our study. Although one might expect a much sparser solution when using such a strict penalty with a comparably large penalty weight, one can see that the respective solution is less interpretable and does not contain more zero elements than the solution favored by the BIC. In fact, almost all parameter values are shrunk which leads to a rather uninterpretable solution (i.e., many variables load approximately equally on all three factors). In addition, one loading (i.e., the loading of Item 13 on the third factor) is estimated to be greater than one which obviously makes no sense for standardized parameters and can be compared to cases where an inadmissible factor rotation is performed (e.g., Jennrich, 1979). This example illustrates that researchers are most likely not able to set γ to a specific value based on a theoretical assumption and expect the method to perform well, but that they should choose it in a data-driven way instead.

Table 2. Comparing estimated loading matrix for data example between BIC-favored (“best”) solution and solution with largest penalty weight

	Best solution			Largest penalty		
	Factor 1	Factor 2	Factor 3	Factor 1	Factor 2	Factor 3
Item 1	−0.42	0.24	0.00	−0.19	0.19	0.17
Item 2	−0.61	0.09	0.06	−0.22	0.20	0.18
Item 3	−0.22	0.58	0.02	−0.03	0.27	0.20
Item 4	0.00	0.71	−0.05	0.04	0.25	0.13
Item 5	−0.09	0.32	0.39	0.29	0.23	0.18
Item 6	−0.06	0.52	0.07	0.07	0.21	0.15
Item 7	−0.28	0.03	0.31	0.10	0.15	0.14
Item 8	0.00	0.23	0.64	0.51	0.24	0.16
Item 9	−0.41	0.16	0.00	−0.19	0.16	0.14
Item 10	−0.22	0.24	0.34	0.17	0.23	0.17
Item 11	−0.35	0.29	−0.10	−0.22	0.16	0.15
Item 12	−0.04	0.10	0.64	0.47	0.20	0.15
Item 13	0.04	0.80	0.00	0.00	−0.87	1.17
Item 14	−0.60	0.00	0.06	−0.22	0.17	0.16
Item 15	−0.29	0.32	0.14	0.00	0.23	0.17
Item 16	−0.11	0.00	0.57	0.36	0.16	0.14

If researchers do not want to choose between the LASSO and Ridge penalty, they can select the ElasticNet penalty instead and choose the mixture between LASSO and Ridge penalty also in a data-driven way via BIC. This appears to be meaningful since the “best” LASSO model does not provide a very sparse solution and we might have to assume that the data actually calls for a factor solution with substantial cross-loadings which is also in line with the model proposed by Fassbender and Luhmann (2021) (i.e., a model with a general factor and three residual factors).

```
# apply the cv_regsem function with the same settings as
# before, only changing the type of penalty and using
# different alpha values that determine the mixture of the
# LASSO and Ridge penalty terms
# grid for alpha values ranging from 0 to 1 with 0.05 as
# step-size
# fitting the model repeatedly for several different values of
# alpha and the penalty weight gamma (here: lambda)
# can take quite a while depending on model complexity
# and computer hardware
# users might want to consider fewer values for both
# hyperparameters
```

```
results_both_alpha = lapply(seq(0, 1, by = 0.05),
function(i){
  cv_regsem(model = regsem_mod, pars_pen =
c(Loadings, Cors),
  mult.start = TRUE, multi.iter = 10, metric =
“BIC”,
  n.lambda = 200, type = “enet”, jump = 10−5,
  lambda.start = 0.0001, alpha = i)
})
```

```
# determine alpha via BIC automatically:
```

```
index = which.min(unlist(lapply(1:length(
seq(0, 1, by = 0.05)),
FUN = function(i){
  min(results_both_alpha[[i]]$fits[, 4])
})))
alpha = seq(0, 1, by = 0.05)[index]
# alpha = 0.45 provides best results

# extract parameter estimates for best hyperparameter
# set according to BIC

model_pars =
results_both_alpha[[index]]$final_pars
```

For the exemplary data, the BIC indeed favors the solution for $\alpha = .45$ which means that the LASSO and Ridge penalty are almost equally weighted. However, when looking at the actual BIC values of the different solutions, the LASSO penalty ($\alpha = 0$) yields almost the same BIC ($BIC_{LASSO} = 18,470.13$ compared to $BIC_{0.45} = 18,470.12$). Using the Ridge penalty, though, the model fit is comparably worse according to the BIC ($BIC_{RIDGE} = 18,506.67$). Hence, the presented results using the LASSO penalty are arguably the most promising when using REFA. Researchers who are used to the common two-step approach still might want to compare REFA to the “traditional” way of conducting an EFA. Again, using the two-step approach with Oblimin rotation, we end up with quite similar patterns compared to our results presented in Table 2.

The *penfa* Package

An alternative to the *regsem* package, *penfa* (Geminiani et al., 2021a) was introduced by Geminiani and colleagues (2021b). Contrary to *regsem*, *penfa* offers the MC+, SCAD, and adaptive LASSO penalties, while the *ElasticNet* penalty is not provided. A nice feature of *penfa* is the automatic hyperparameter selection based on a trust-region optimization algorithm minimizing an approximate AIC (see Geminiani et al., 2021b for more details) which promises a faster and better selection of the penalty weight. However, this automatic procedure is only available for the Ridge, LASSO, and adaptive LASSO penalty, while researchers still need to specify a grid of potential values for other penalties such as the MC+ penalty. In the following, we will provide exemplary code that showcases how *penfa* can be used for REFA.

```
library(penfa) # load package

# Model specification using typical lavaan syntax
## ... is a replacement for items 3 to 15, users who want to
## actually run the model need to replace it with pros3 +
## pros4 + pros5 and so on
## the full model syntax is presented in the supplemental
## material
## we use ... to save space here!

model = "F1 =~ pros1 + pros2 + ... + pros16
        F2 =~ pros1 + pros2 + ... + pros16
        F3 =~ pros1 + pros2 + ... + pros16
        F1 ~~ F2
        F1 ~~ F3
        F2 ~~ F3
        F1 ~~ 1*F1
        F2 ~~ 1*F2
        F3 ~~ 1*F3
        "
```

Applying grid search to determine the penalty weight is less convenient with *penfa* compared to *regsem*. The *penfa* function takes a specific penalty weight and performs penalized maximum likelihood estimation using this exact penalty. It does not accept a vector of different values, so users who want to try out different values and chose the penalty weight using the BIC, for example, need to loop through all values of their prespecified grid. Hence, the respective code becomes a little more complicated. However, as mentioned above, *penfa* provides an option for automatic parameter selection when using the LASSO penalty.

Usually, the automatic procedure is faster than a grid search approach and does not require the user to specify

```
# grid of potential penalty weights

grid = seq(.0001, by = 0.00001, length.out = 200)

# function that conducts penfa given a specific penalty
# weight

# note that the same penalty weight is used for loadings and
# factor correlations
# latent and observed variables are standardized for EFA
# LASSO penalty is chosen and strategy = "fixed" as grid
# search is applied

penfa_lasso = function(pen){
  penfa(model = model, data = data, pen.shrink =
    "lasso", strategy = "fixed",
    std.ov = TRUE, std.lv = TRUE, information =
    "hessian",
    eta = list(shrink = c("lambda" = pen,
      "phi" = pen),
    diff = c("none" = 0)))
}

# apply penfa with lasso penalty using different penalty
# weights

results = lapply(grid, penfa_lasso)

# calculate BIC for each solution
# (tryCatch function for exception/error handling when a
# solution
# for a given penalty weight was inadmissible)

bics = sapply(results,
  function(x) {tryCatch(BIC(x),
    error = function(e) {return(NA)}})})

# determine optimal penalty weight according to BIC

pen.weight = grid[which.min(bics)]

# extract and display loading matrix for "optimal
# solution"

penfa_pars =
  round(matrix(results[which.min(bics)][[1]]
    @ParTable$est[1:48],
    ncol = 3), 2)
```

a grid of potential values for the penalty weight. Accordingly, it avoids a scenario where researchers specify a grid for the penalty weight that is too narrow or only includes values that are too small (i.e., penalizing too little) or too large (i.e., penalizing too strongly). Nonetheless, Geminiani and colleagues (2021b) point out that the automatic procedure as implemented could face difficulties in small sample conditions from time to time – while its overall performance in their simulation study was very good.

The *IsIx* Package

```
library(IsIx) # load package

# Model specification using IsIx syntax
# last three rows impose unit variance constraints
# so that latent variables are standardized

## ...is a replacement for items 3 to 15, users who
## want to actually run the model need to
## replace it with pros3 + pros4 + pros5 and so on
## the full model syntax is presented in the
## supplemental material
## we use ... to save space here!

model_IsIx = "F1 =~> pros1 + pros2 + ... + pros16
              F2 =~> pros1 + pros2 + ... + pros16
              F3 =~> pros1 + pros2 + ... + pros16
              F1 <=~> F2
              F1 <=~> F3
              F2 <=~> F3
              F1 <=~> 1*F1
              F2 <=~> 1*F2
              F3 <=~> 1*F3"
```

Another alternative is the *IsIx* package by Huang (2020a). Since it has been proposed for (semi-)confirmatory SEM, users who want to use it for REFA need to specifically address the standardization of the observed variables. *IsIx* works with its own syntax (see above) that is similar to the well-known *lavaan* syntax. A nice advantage of *IsIx* is that one can specify the exact parameters that should be penalized directly within the syntax (which may come in handy if you have a semi-confirmatory setting where you only want to penalize cross-loadings or selected between-factor correlations, etc.). In our example, since we are conducting a fully exploratory analysis, we penalize all factor loadings and between-factor correlations.

```
# initialize new model
# scale data so that observed variables are standardized

IsIx_efa = IsIx$new(model = model_IsIx, data =
  scale(data))

# fit the REFA model with elastic net penalty

IsIx_efa$fit(penalty_method = "elastic_net",
  lambda_grid = seq(.0001, by = 0.00001,
    length.out = 200),
  delta_grid = seq(0, 1, y = 0.05))

# fit the REFA model with lasso penalty
# (either do one or the other)

IsIx_efa$fit(penalty_method = "lasso",
```

```
lambda_grid = seq(.0001, by = 0.00001,
  length.out = 200))

# select penalty weight (and elastic net hyperparameter,
# here delta) using BIC

IsIx_pars = matrix(IsIx_efa$extract_coefficient(selector = "bic") [17:64],
  ncol = 3)
```

Contrary to *regsem*, *IsIx* allows researchers to fit REFA with ElasticNet penalty and different weights γ (internally called delta, see `delta_grid` argument) in one go without having them to loop through all values provided in a grid (when using *regsem* we had to *lapply* to try out different values of γ). Another difference for users is that *IsIx* is an R6 class, that is, it is based on an object-oriented programming system which changes the way models are fitted (fitting a model is a functionality of the *IsIx* object itself, see code above) as well as how results are retrieved. The general modeling idea, however, remains the same.

Comparison of REFA Implementations

All three packages have their advantages and disadvantages, so users need to find out which functionalities (e.g., which penalties) they need, and which implementation provides those in the most accessible manner. Orzek and colleagues (2023) demonstrate that the estimation algorithm (relying on a non-specialized global purpose optimizer) implemented in *regsem* in combination with model selection via BIC may yield unstable results. In addition, *regsem* is not the most efficient implementation taking quite a long time for parameter estimation and needing more computational resources when the grid of evaluated γ values becomes larger and the sample size, as well as the number of model parameters, are increased (i.e., more latent and observed variables). Accordingly, for comparably complex models, the optimization routine implemented in *IsIx* appears to be preferable when relying on the LASSO penalty (Huang, 2018).

However, as *IsIx* is primarily designed for (semi-)confirmatory approaches, using it for EFA requires users to preprocess the data (standardization of manifest variables) and specify the latent variance (standardization of latent variables) if they want to obtain standardized factor loadings. Since inexperienced users are probably a bit insecure when it comes to specifying a range of potential penalty values, the automated optimization approach implemented in *penfa* appears to be quite appealing when using LASSO, Ridge, or an adaptive LASSO penalty (Geminiani et al., 2021a). In addition, *penfa* (as well as *regsem*) uses several

routines from *lavaan* (Rosseel, 2012) which most users are familiar with, which might be an advantage over *lsx*. Contrary to *regsem* which is focused on standard SEM, *penfa* and *lsx* also cover multi-group cases (e.g., Huang, 2018) which are especially of interest if between-group differences are assessed and measurement invariance has to be established (e.g., De Roover, 2021; French & Finch, 2008; van de Schoot et al., 2012). *lsx* is arguably the most versatile implementation as it does not focus on simple factor models as *penfa*, enables researchers to use various different penalty functions, and is computationally more efficient (i.e., considerably faster) than *regsem* which arguably makes it the state-of-the-art implementation of frequentist REFA.

There are also Bayesian regularization methods for EFA (e.g., Carvalho et al., 2008; Chen, 2021) that achieves sparsity through shrinkage priors instead of penalty functions. Van Erp (2023) put together a nice introduction to the different software solutions that can be used for (regularized) Bayesian SEM (not a particular focus on EFA). The overview also provides exemplary code, so that users who want to try out Bayesian regularization are referred to her work.

When (Not) to Use Regularized EFA

As Scharf and Nestler (2019) showed (and our first example illustrated), REFA is a reliable alternative to the common two-step approach of EFA with subsequent factor rotation. Using REFA does not necessarily yield better results compared to standard EFA, but promises to leave out ambiguous decisions with regard to different rotation techniques. In addition, Jacobucci and colleagues (2016) show that regularized SEM usually provides solutions that are more generalizable to new data than the common counterparts. Yarkoni and Westfall (2017) explain how regularization is used in machine learning to obtain models that generalize well trading bias in parameter estimation against smaller estimation variance – which is known as the bias-variance trade-off (see also Hilbert et al., 2021 for a gentle introduction). When comparing the two-step EFA approach to REFA, researchers should be aware that the in-sample model fit of REFA will be slightly worse compared to EFA with factor rotation (Jin et al., 2018). Even though the reduced model fit appears to be a severe downside of REFA, penalizing the number of parameters and therefore shrinking parameter values to zero promises better out-of-sample fit, that is, better generalizability of the model to new data. Since EFA is usually used to explore the latent structure of a psychological concept, and the results are usually validated with confirmatory analyses in new samples, generalizability plays a key role EFA-based research. It has to be stated though, that differences between common EFA with an appropriate rotation method and REFA

will be very small in common psychological application contexts, REFA's full potential will probably come into play with larger item sets and rather sparse loading patterns (e.g., Hirose & Yamamoto, 2015). In ESM 2, interested readers can find a data example where different strategies are compared with regard to out-of-sample performance. It can be seen that for data that is typical in psychology (e.g., Big 5 personality measures), differences are rather small – however, REFA does yield slightly better-fitting confirmatory factor analysis (CFA) models on new data.

In general, the framework of regularized SEM is even more versatile. Instead of penalizing all loading parameters, researchers might want to opt for a semi-confirmatory approach as suggested by Huang and colleagues (2017) where only cross-loadings are penalized. This way researchers can start with fully exploratory analyses and then penalize selected parameters in subsequent (rather) confirmatory analyses without jumping from an EFA to a CFA with an independent cluster assumption that will almost inevitably fail to fit new data (Hopwood & Donnellan, 2010). A semi-confirmatory approach – where only (some) cross-loadings are penalized and other parameters are freely estimated – can be compared to a rotation-to-target approach (Myers et al., 2015) which is also used in situations where researchers want to incorporate previous study results or theoretical constraints. REFA therefore can be described or rather used as a hybrid (semi-confirmatory approach) between EFA and CFA and therefore serves similar purposes as exploratory SEM (Asparouhov & Muthén, 2009).

When considering regularization for strictly exploratory approaches (i.e., REFA in a narrow sense where all loading parameters are penalized), its results will not differ strongly from the common two-stage approach in EFA (e.g., Scharf & Nestler, 2019). However, given a regularization method (e.g., LASSO), an “optimal” level of sparsity in the factor loading matrix can be achieved by selecting the penalty weight based on an information criterion.

If researchers are interested in confirming hypothesized model structures, they usually choose confirmatory approaches. As indicated above, regularized SEM (e.g., Jacobucci et al., 2016, 2019) is a general framework that can be used for both exploratory and confirmatory analyses. However, researchers should be aware that if they want to conduct statistical inference on specific model parameters, it can be quite challenging to obtain unbiased standard error estimates in regularized models (e.g., Huang, 2020b). Since this paper focuses on a completely exploratory methodology, we will not discuss this issue more thoroughly. Interested readers can find a general discussion of statistical inference in the context of regularization and variable selection at Kuchibhotla and colleagues (2022). Huang (2020b) focuses on this issue more thoroughly in the context of SEM and factor analytic approaches.

Conclusion

Since finding the most suitable rotation method can be a challenging task for researchers due to the lack of a data-driven way to determine the best factor rotation, that is, due to the problem of rotational indeterminacy (Browne, 2001; Schmitt & Sass, 2011), REFA seems to be a good alternative to the common two-step approach in EFA. By adding a penalty term to the log-likelihood function, the issue of rotational indeterminacy is resolved. In fact, REFA produces unique factor structures up to factor ordering and sign switches (see also Scharf & Nestler, 2019). Therefore, we believe that REFA should be considered a meaningful alternative for users of EFA more often. Arguably, REFA is not only associated with less ambiguity but might also be favorable as the level of sparsity in the loading pattern can actually be determined in a data-driven way. However, users should keep in mind that REFA results also need to be thoroughly cross-validated on new data (i.e., confirming the measurement model using a CFA) and therefore need to be reported transparently.

With this paper, we want to encourage researchers in psychology and psychological assessment to add REFA to their toolbox. Most certainly, REFA will not solve all problems and ambiguities that are associated with classical EFA and factor rotation in particular, but it allows to development of measurement models in a first step in a more data-driven and maybe less subjective way.

Electronic Supplementary Materials

The electronic supplementary material is available with the online version of the article at <https://doi.org/10.1027/1015-5759/a000792>

ESM 1. Code to run Regularized EFA with smaller grid for lambda

ESM 2. Comparing classical EFA and REFA on out-of-sample data

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Open Science

We report how we determined our sample size, all data exclusions (if any), all data inclusion/exclusion criteria, whether inclusion/exclusion criteria were established prior to data analysis, all measures in the study, and all analyses including all tested models. If we use inferential tests, we report exact *p* values, effect sizes, and 95% confidence or credible intervals.

Open Data: The data used for the exemplary analysis can be downloaded from the OSF repository (<https://osf.io/rp8au>) of the original study by Fassbender and Luhmann (2021).

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